A Deep Ensemble-Driven Framework Integrating Radiomics and Segmentation-Based Attention Mechanisms for Brain Abnormality Localization and Classification

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Abstract: Despite the variability of the appearance of abnormal tissue and the imbalance of classes, brain tumor segmentation is a very difficult task in magnetic resonance imaging. This study aims to explore and evaluate a deep learning architecture that consists of a lightweight SegNet encoder–decoder model and a residual U-Net model, along with exploratory ensemble prediction and radiomics analysis. The dataset for the segmentation experiment is taken from a BraTS2020-derived dataset and consists of 2000 slices, divided into 1400 training slices, 300 validation slices, and 300 test slices. Images of four channels are resized and standardized using z-score normalization, and supervised by the first channel of the mask. Both networks are optimized with Adam and trained for 50 epochs with a combined objective consisting of 60% Dice loss and 40% binary cross-entropy. Optimization: Gradient clipping, and learning-rate reduction based on validation data. The highest final validation Dice score of SegNet and U-Net is 0.8461 and 0.8793 respectively, and recorded final Dice scores are 0.8449 and 0.8758 respectively. Equal weight probability averaging is used, but not quantified for segmentation. Validated classification claims are explored with radiomics extraction and fully connected classification, but there are inconsistencies. There is a separate custom U-Net interface that displays segmentation overlays and Grad-CAM visualization. The contribution is based on an implementation-focused comparative framework, with a limited conclusion about patient-level generalization due to the splitting of slices and the lack of a held-out test evaluation. For further evaluation, results must be consistently preprocessed and independently confirmed.

Keywords: Brain tumor segmentation, magnetic resonance imaging, U-Net, SegNet, ensemble learning, radiomics, Grad-CAM.

I. Introduction

Brain tumor segmentation is a computational problem in Magnetic Resonance Imaging (MRI) to locate areas of abnormality in brain images. The difficulty of automated delineation is caused by the heterogeneous appearance of tumors in different slices, the irregular contours and the fact that abnormally small parts of the image are occupied by tumors. Unlike other imaging channels, intensities of the difference images vary from one image to the next, further complicating interpretation of features. These properties highlight the need for segmentation methods that use spatial representation learning methods that are also foreground-overlapping objective-sensitive, as opposed to pixelwise classification.

Hierarchical representation learning and spatial prediction mapping are demonstrated by encoder-decoder neural networks, which make this task a practical possibility. U-Net adds skip connections between the encoding and decoding layers, so that the output of the reconstruction can include features at multiple resolutions. This project is based on the SegNet-style lightweight network which utilizes convolutional encoding, batch normalization, pooling and bilinear upsampling. These architectures are investigated under a common training setup, which allows for a comparison of the segmentation behavior while keeping the following aspects fixed: data preparation, optimization parameters, and the evaluation method.

In this work, we look at the four-channel MRI slices of a set of data derived from BraTS2020. A filtered subset consists of 2000 tumor slices which are divided into 1400 training, 300 validation, and 300 test sets. Images are resized to 256 × 256 pixels and then normalized using z score normalization. The three channels of the source masks are used but only the first channel is used in the implemented segmentation target. Therefore, the experiment does not aim to achieve simultaneous segmentation of all the identified tumor parts, but only at delineating a single channel.

Both segmentation networks are optimized with a weighted combination of Dice loss and binary cross-

entropy, clipped gradients, and adaptive learning-rate reduction. The notebook also uses equal weight averaging of model probabilities. This operation offers a simple ensemble mechanism, but there is no quantitative ensemble evaluation. An exploratory radiomics branch is used to extract the numerical descriptors and determine a fully connected classifier. Due to the differences in the dimensions of features and the availability of the labels and saved artifacts it is not possible to treat the classification output from this as a verified experimental result.

The accompanying application provides the same performance as the above, but with a different custom U-Net to create segmentation overlays and Grad-CAM visualizations from uploaded images. This interface has interactive inference, but it's different from the residual U-Net used in the notebook. The study thus focuses on repeatability of architectural description and unambiguous interpretation of validated results. It is limited in scope by the use of a slice-level partitioning, the fact that it is not reported that the classification was reproducible and the absence of an evaluation of the reserved test subset. These are used to separate out implemented functionality from experimentally determined performance and to set the bar for further validation and larger claims for generalization.

Contributions

- Training SegNet-style and residual U-Net models with the same MRI segmentation setup.
- Comparative reporting of recorded segmentation losses and validation Dice.
- Implementation of probability-averaging fusion, exploratory radiomics-based classification components.
- Implementation of a custom U-Net interface for segmentation overlays and visualization using Grad-CAM.

Novelty

One unique feature of this work is the integration of comparative segmentation, probability level fusion, exploratory radiomics, and interactive visualization in the same project, all with the goal of being implemented in the work. The contribution is to the organization and examination of these components, not a new neural architecture or a new learned stacking algorithm. Architectural comparison and visualization functionality is supported by the available evidence, whereas ensemble improvement and radiomics classification are not. Thus, novelty is presented as a contribution to the application level integration that needs to be further evaluated and compared with existing published frameworks in a reproducible way.

II. Literature Survey

This recent medical imaging research paper focused on attention mechanisms, hybrid architectures, and radiomics for characterization of lesions. Wang et al. [1] studied the world-wide attention on multi-sequence MRI for early diagnoses of femoral head osteonecrosis. Himel et al. [2] conducted a review of deep learning applications in MRI of osteosarcoma. These studies provide background information on MRI analysis but their anatomical focus is not on tumor of the brain.

Liu et al. [3] have looked at deep learning segmentation and dual-phase CT radiomics to distinguish lipid-poor adrenal adenomas from other lesions of the adrenal glands. Wang et al. [4] studied the radiomics of the hippocampus subfield and its association with inflammation and cognitive impairment in chronic kidney disease. These are relevant not to segmentation of gliomas, but to quantitative image characterization.

Behzadpour et al. [5] proposed an improved ResU-Net that uses EfficientNetB0, channel attention and atrous spatial pyramid pooling for brain tumor segmentation. These components deal with feature extraction and multiscale representation. Aresta et al. [6] reviewed the techniques for segmentation and classification of gliomas from multiparametric MRI. It is important to note that their related preprint [12] has the same research theme and cannot be regarded as independent experimental evidence.

A two-stage geometric-attention approach to lung nodule segmentation and uncertainty-aware malignancy classification was proposed by Sufyan et al. [7] in federated settings. Sowmiya and Fatima [8] introduced a segmentation guided CNN–vision transformer architecture for explainable detection and classification of brain tumors. Ramamoorthy et al. [9] introduced TriMedNet, which combines imaging, medical text and numeric patient data. These methods highlight various feature learning – downstream diagnostic tasks connections.

Lu et al. [10] introduced a three-dimensional Transformer–CNN structure with Gaussian gating, layered refinement, and an edge–region weighted objective, which is named GL-Net. Seenu et al. [11] proposed MANet to make a classification of brain tumor using multimodal attention. These researches separate the spatial delineation from image classification, and confirm the importance of the assessment of tasks with proper targets and evaluation measures.

A hybrid three-dimensional vision transformer and U-Net model is provided in the supplied UFTrans reference [13] but detailed experimental claims are not presented as it is the primary publication that was not retrieved. Padia and Meva [14] designed the cross-attention and adaptive fusion for fetal ventriculomegaly detection and grading. Saifullah et al.

[15] used U-Net attention mechanism and PSO in brain tumor segmentation.

Altogether, these publications inspire the discussion on representation learning, fusion, quantitative descriptors, and interpretability. Differences in anatomy, modalities, targets for segmentation and validation protocols, however, make it difficult to make direct comparisons of the performance. Comparative segmentation and exploratory integration are therefore the focus of the present project, which does not assert architectural originality or superiority over comparative segmentation and exploratory integration.

Research Gap

The reviewed literature justifies a methodological question: Will the combination of segmentation output enhance delineation and generate reliable radiomic descriptors of the process of consistent preprocessing and independent evaluation. This has not been resolved for the present project for two reasons: (1) ensemble performance is not reported and (2) feature–label alignment is incomplete and validation is done on a slice level partition. The present study therefore focuses on comparative segmentation, as a first step towards integration. More robust findings await until patient-separated testing, validated radiomics extraction, component ablation, and the alignment of experimental models and deployed inference are established first

III. Proposed Methodology

A. Dataset Preparation and Image Preprocessing

The experimental setup consists of four-channel MRI slices of a dataset from the BraTS2020 challenge in HDF5 format. On the source mask channels, a filtering procedure is used to keep 2000 slices with nonzero annotations. Random partitioning is done with the random state 42, partitioning 1400 slices into training, 300 into validation, and 300 into testing. This partition works on the slice level without enforcing any separation between patients. All images are resized from 240 × 240 to 256 × 256 pixels, using linear interpolation; the masks are resized using nearest-neighbor interpolation. Image intensities will be globally z-scored:

$$\hat{X} = \frac{X - \mu_X}{\sigma_X} \qquad (1)$$

Here, $\mu_X$and $\sigma_X$represent the image mean and standard deviation. Normalization is bypassed when the standard deviation is below 10^{-5}. Only the first mask channel supplies segmentation supervision.

B. Segmentation Network Construction

The selection of the segmentation target is learned by two networks independently. It is a SegNet style encoder with 64 and 128 filters, batch normalization, ReLU activations, and max pooling, and is lighter than others. It's decoder recovers spatial resolution using bilinear upsampling and convolution. The residual MONAI U-Net is designed with 16, 32, 64 and 128 channels and 3 downsampling strides and 1 residual unit. Four input channels are supported and four output channels are provided, with both of them providing logits in only one channel. Their probabilities are given in terms of probability maps as:

$$P_s = \sigma\left(f_s(\hat{X};\theta_s)\right),\ P_u = \sigma\left(f_u(\hat{X};\theta_u)\right) \qquad (2)$$

Here, $f_s$and $f_u$denote the segmentation networks, while $\sigma$is the sigmoid function. Parameters are optimized independently.

C. Training Objective and Optimization

Training proceeds for 50 epochs with batch size eight and Adam optimization at an initial learning rate of 10^{-4}. The objective combines foreground overlap and pixelwise prediction penalties:

$$L = 0.6\left(1 - \frac{2\sum_i p_i\, y_i + 1}{\sum_i p_i + \sum_i y_i + 1}\right) + 0.4 L_{\mathrm{BCE}}(P, Y) \qquad (3)$$

Here, $p_i$and $y_i$denote predicted probabilities and target values; overlap sums cover the batch. Gradient clipping limits the parameter-gradient norm to 1.0. A validation-loss scheduler reduces the learning rate by a factor of 0.1 after five non-improving epochs. The implementation shares one checkpoint-selection minimum between networks, limiting independent best-model selection.

D. Ensemble Prediction and Radiomics Exploration

The later ensemble implementation averages sigmoid probability maps and applies binary thresholding:

$$\widehat{M} = \mathbb{1}\left[\frac{P_s + P_u}{2} > 0.5\right] \qquad (4)$$

This is a form of probability averaging and not learned stacking. It has no reusing of the training preprocessing in its inference process and performance of ensemble has not been quantified. PyRadiomics and SimpleITK extract descriptors from training images and masks, not ensemble masks. This exported table has 11 features (diagnostic and shape). A fully-connected classifier has hidden layers of 128 and 64 neurons as well as 2 logits for output. The conversion from channel to spatial axis, however, makes the evaluation of classification repeatable, but the lack of labels and irregularity of feature dimensions make it difficult to repeat the classification evaluation.

E. Evaluation and Interactive Visualization

Segmentation evaluation thresholds probabilities at 0.5 and computes batchwise Dice overlap:

$$D = \frac{2\sum_i \widehat{m}_i\, y_i}{\sum_i \widehat{m}_i + \sum_i y_i + 10^{-6}} \qquad (5)$$

Averages scores are calculated over validation batches. Reserved test subset - No evaluation reported. A Streamlit application is developed to process grayscale images with the custom U-Net at a resolution of 512 × 512, independently. It shows segmentation overlays, Grad-CAM heatmaps generated by back-propagating the summed segmentation output from the final convolution. This deployed model is different from the one used in the notebook's MONAI U-net, so validation scores are not recorded for the application.

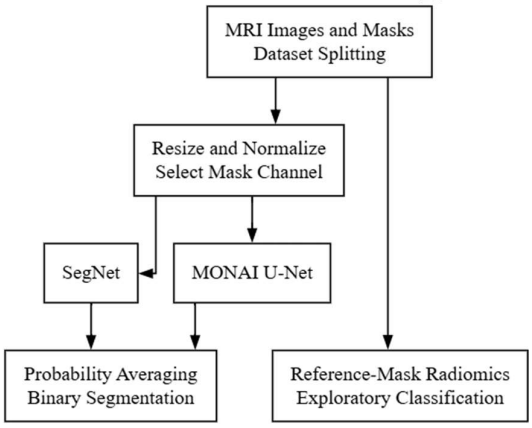


Figure 1. Architecture Diagram

IV. Result and Discussion

A total of 1,400 MRI slices with tumor samples were used for training, 300 for validation, and 300 for testing in a segmentation experiment. Both SegNet and the residual MONAI U-Net were optimized for 50 epochs with the same optimization configuration. The results available include saved training and validation logs; evaluation on the reserved test subset was not reported. Thus, findings should not be taken as patient-level performance without additional independent validation within the slice-level partition implemented.

As detailed in Table I, for the final validation step, U-Net scored 0.8758 in Dice whereas SegNet scored 0.8449. The absolute difference in this case is 0.0309 or 3.09 percentage points. U-Net also achieved the lowest validation loss of 0.0806 while SegNet achieved 0.0979. Under the recorded experimental conditions, these observations suggest that the performance of U-Net is closer to the chosen reference-mask channel. They are not generalizable to other data sets, tumor components, or patient populations.

TABLE I. RECORDED SEGMENTATION PERFORMANCE AT EPOCH 50

Metric	SegNet	MONAI U-Net
Training loss	0.0714	0.0379
Training Dice	0.8847	0.9479
Validation loss	0.0979	0.0806
Validation Dice	0.8449	0.8758

The results of the training show another difference between the networks. The training Dice score of U-Net was 0.9479, higher than SegNet's 0.8847. It has a Dice score of 0.0721 when trained and validated against the test set, whereas SegNet has a Dice score of 0.0398. The bigger the gap, the better the fitting of training samples and the gap merits further appraisal of generalization. It does not prove overfitting independently since the resulting scores are based on the overlap of a single partition and add up to the batch level. However, difference is adequate for assessing model behavior through patient separated subsets and repeated experiments.

Table II differentiates between minimum validation loss and maximum validation overlap. The best validation Dice and lowest validation loss of SegNet were obtained at epoch 11 with a score of 0.8461 and 0.0973 respectively. The best U-Net obtained was a best Dice of 0.8793 with the epoch number of 41, and the best validation loss was 0.0805 at epoch 46. It had a minimum-loss epoch, and at this time, its Dice score was 0.8780. Choosing checkpoints with the combined loss function does not necessarily correspond to the epoch having the largest number of points above the threshold.

TABLE II. VALIDATION OPTIMA IDENTIFIED FROM SAVED TRAINING LOGS

Validation criterion	SegNet	MONAI U-Net
Highest Dice	0.8461	0.8793
Epoch of highest Dice	11	41
Lowest loss	0.0973	0.0805
Epoch of lowest loss	11	46
Dice at lowest-loss epoch	0.8461	0.8780

The notebook provides no ensemble Dice or loss measurements, only implements probability averaging. So it can't be said that there is improvement through fusion. Additionally, ensemble inference skips over the training pre-processing procedure, and creates an undecided consistency issue. The radiomics branch is also still in its infancy: the feature table provided has 100 samples and 11 features, while the classifier saved requires 21 inputs. The 100% classification accuracy displayed is not validated as it is not clear whether classification labels are missing or notebook states are inconsistent.

Qualitative notebook outputs include outputs for predicted masks, segmentation overlays, and outputs for the Grad-CAM visualisation. The app also showcases image upload and custom U-Net inference, but with a different architecture than the one tested in the MONAI model. These visualizations represent functionality, but do not determine the accuracy of localization or the fidelity of explanation. Additionally, two constraints limit interpretation: first channel mask supervision and the presence of tumors in the slice selection and potential patient overlap between partitions. In general, the evidence in this experiment favors U-Net, and the effectiveness of ensemble, reproducible radiomics classification and independent generalization are not confirmed.

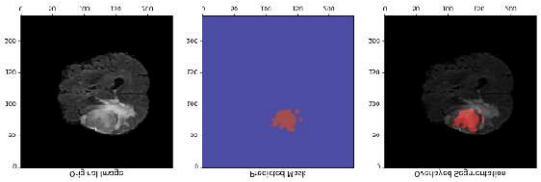


Fig. 2. SegNet segmentation example showing the original MRI slice, predicted mask, and segmentation overlay.

We show the MRI slice in Fig. 2, SegNet prediction, and the overlay in the same figure. The area marked in yellow represents the predicted area of foreground localization; it is impossible to determine visually whether this area is correct or incorrect without comparing it to the reference mask.

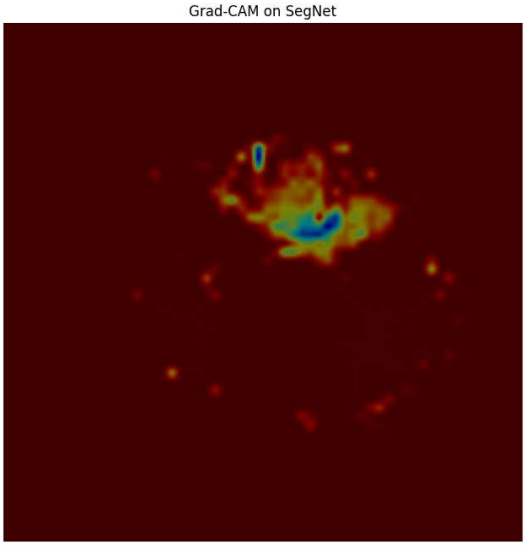


Fig. 3. Grad-CAM visualization of SegNet for a validation MRI slice.

We present an illustration of the spatial contributions to SegNet's overall segmentation output using Grad-CAM in Fig. 3. The heatmap is used to give a qualitative understanding of the model's answer, and is not used to independently validate the location of the tumor or the confidence of the explanation.

V. Conclusion and Future Scope

In this work, we evaluated the segmentation of brain MRI with a lightweight SegNet-inspired network and a residual MONAI U-Net as well as exploratory ensemble prediction and radiomics analysis. The final validation Dice scores were 0.8758 for U-Net and 0.8449 for SegNet, showing that the overlap between the two models was better in the experiment implemented. Probability averaging and radiomics classification were used, but the effectiveness has yet to be verified. A custom U-Net application was used to display segmentation overlays and Grad-CAM visualization in a separate application. These results can be used for comparative segmentation analysis, but cannot be used for patient-level generalization or clinical effectiveness. Patient-separated partitioning, consistent preprocessing, and evaluation of the reserved test subset will be a focus of future work. Ensemble predictions should be compared quantitatively and result in prediction component ablation. To process radiomics extraction, spatial interpretation needs to be corrected, the correspondence between samples and labels needs to be maintained and the correspondence between radiomics features being exported and the radiomics features being used as an input to the classifier needs to be preserved. Other priorities are repeatability of the selection of checkpoints, incorporation of validated models in the application, and measuring the fidelity of the explanation. In addition, external validation would provide robustness under imaging conditions.

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